

Python for Agentic AI Engineering

Complete Study Notes

Student Name: _____

Week 0 - Setup & First Run

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1 Week 0: Setup & First Run

1.1 Class 1: What is Programming?

Key Concept 1: Computer Kya Hai?

- **Computer:** Ek powerful brain hai jo instructions follow karta hai
- Computer **khud kuch nahi samajhta** - usey har kaam ke liye **step-by-step instructions** deni padti hai
- **Analogy:** Jaise ek chhota bachcha - agar aap kehte hain "paani lao" to bachcha galat tareeqa istemal karega, lekin agar step-by-step batayen to sahi karega

Key Concept 2: Programming Language

- **Programming Language** = Zariya jis se hum computer ko instructions dete hain
- **Computer Ki Language** = Sirf **0 aur 1** (Binary Language)
 - 0 = No electricity / Current nahi hai
 - 1 = Electricity hai / Current hai
- **Machine Level Language** = 0 aur 1 mein likhi gayi language (sirf machine samajhti hai)
- **High Level Language** = Insaan ko samajh aane wali language (jaise Python, Java, C++)

Key Concept 3: Python

- **Creator:** Guido van Rossum (1991 mein banayi)
- **Features:**
 - Simple syntax - English jaisi lagti hai
 - Beginner friendly - Easy to learn
 - Aaj kal sab se zyada popular language

- **Example:**

```
1 # Python Code (Insaan samajh sakta hai)
2 if marks >= 35:
3     print("Pass")
4 else:
5     print("Fail")
```

1.2 Class 2: Interpreter vs Compiler

Interpreter (Python)

- **Line-by-line** code execute karta hai
- Jaise khana **ek-ek item** khaana (pehle Manchurian, phir Dahi Vada)
- Agar ek line mein error ho to wahan ruk jaata hai
- **Python, JavaScript, Ruby** mein use hota hai

Compiler

- **Poora code ek saath** compile karta hai
- Jaise **thali mein sab kuch** le kar khana
- Phir execute karta hai
- **C, C++, Java** mein use hota hai

Comparison Table

Feature	Interpreter	Compiler
Execution	Line-by-line	All at once
Speed	Slower	Faster
Error Detection	Stops at error	Shows all errors
Memory Usage	Less	More
Languages	Python, JS	C, C++, Java

1.3 Class 3: Agentic AI Engineering

What is Agentic AI?

- **Agent** = Ek system jo soch sakta hai aur kaam kar sakta hai
- **3 Components:**
 1. **Think** (LLM - Large Language Model): Sochne ke liye brain
 2. **Tools**: Kaam karne ke liye instruments
 3. **Loop**: Jab tak kaam na ho, try karte rahna
- **Real World Example:**
 - Task: Paani lana
 - Think: Paani kahan hai?
 - Tools: Paanv (chalo), Haath (pakdo), Glass (bharein)
 - Loop: Agar glass toot gaya → dusra glass lao → dubara try karo

2 Week 0: Environment Setup

2.1 Class 4: Google Colab Setup

Google Colab

- **Online Python Editor** - browser mein chalta hai
- **Kisi bhi device** se use kar sakte hain (Mobile, Tablet, Laptop)
- **Free Resources:**
 - 15 GB free storage (Google Drive)
 - 12.6 GB RAM
 - 107 GB Storage

How to Use Google Colab

1. Google Chrome kholen
2. Search karen: "Google Colab"
3. Sign in with Google ID
4. "New Notebook" click karein
5. Code likhein aur **Run** karein

Example:

```
1 # Code Cell mein likhein
2 print("Hello Khushi")
3 print("Welcome to Python for Agentic AI Engineers Course")
```

Output:

```
Hello Khushi
Welcome to Python for Agentic AI Engineers Course
```

2.2 Class 5: Installing Python Locally

Python Installation Steps

1. Go to `www.python.org`
2. Click "Download Python"
3. **CRITICAL:** "Add Python.exe to PATH" checkbox **select** **karein**
4. Click "Install Now"
5. Installation complete hone dein

Current Version: Python 3.15 (released January 2026)

- Support till: 2031
- File size: 43.7 MB

Verify Installation

1. Command Prompt (CMD) kholen (Windows + R → cmd)
2. Run these commands:

```
1 # Check version
2 python --version
3 # Output: Python 3.15.x
4
5 # Open Python interpreter
6 python
7 # Now write code
8 print("Hello")
```

2.3 Class 6: Code Editor - Cursor

Cursor Installation

- **Cursor** = Modern code editor (AI-powered)
- Better than VS Code for AI development
- **Installation Steps:**
 1. Go to www.cursor.com
 2. Click "Download for Windows/Mac/Linux"
 3. Install karein (Next → Next → Finish)
- **IMPORTANT:** Beginners ke liye **AI features OFF** rakhein

2.4 Class 7: Terminal Commands

Essential Terminal Commands

Command	Meaning	Example
pwd	Print Working Directory	pwd → /Users/khushi
cd	Change Directory	cd Documents
cd ..	Go back one folder	cd ..
dir (Win) / ls (Mac/Linux)	List folder contents	dir
mkdir	Make Directory	mkdir Python_Project

Example:

```
1 # Current location check
2 pwd
3
4 # New folder banayein
5 mkdir Python_for_AI
6
7 # Folder mein jayein
8 cd Python_for_AI
9
10 # Folder ki contents dikhayein
11 dir
12
13 # Ek folder peeche jaayein
14 cd ..
```

3 Week 1: Your First Program

3.1 Class 8: REPL (Read-Eval-Print-Loop)

What is REPL?

- Python ka **interactive mode**
- **Process:**
 1. **Read** → Code padhna
 2. **Eval** → Code evaluate karna (error check)
 3. **Print** → Output dikhana
 4. **Loop** → Phir se wait karna

Example:

```
1 # Terminal mein 'python' likhne par REPL start hota hai
2 >>> print("Hello Khushi")
3 Hello Khushi
4 >>> 5 + 3
5 8
6 >>>
```

3.2 Class 9: First Python Program (.py file)

Creating a .py File

- Professional programmer **.py files** mein code likhte hain
- **File Extension:** .py (**important hail!**)
- Jaise:
 - .mp3 = Audio file
 - .mp4 = Video file
 - .jpg = Image file
 - .py = Python file

File Naming Rules:

1. **Small letters** use karein
2. **No spaces** - underscore (_) use karein
3. Meaningful name: `first_program.py`, `hello_world.py`

Writing Your First Program

Steps:

1. Cursor kholen
2. Project folder mein Right Click → New File
3. Name: `first_program.py`
4. Code likhein:

```
1 # first_program.py
2 print("Hello Khushi Sharma")
3 print("Welcome to Indian AI Production")
4 print("You are starting Python for Agentic AI Engineer")
```

Run Program:

```
1 python first_program.py
```

Output:

```
Hello Khushi Sharma
Welcome to Indian AI Production
You are starting Python for Agentic AI Engineer
```

4 Practice Exercises

4.1 Exercise 1: Google Colab Practice

Exercise 1

Task: Colab par new notebook banayein aur 3 lines print karein:

1. Apna naam
2. Apna city
3. Dream job

Solution:

```
1 print("Name: Khushi Sharma")
2 print("City: Delhi")
3 print("Dream Job: Agentic AI Engineer")
```

4.2 Exercise 2: Terminal Commands Practice

Exercise 2

Task: Terminal mein yeh commands try karein:

1. Current location check karein
2. Naya folder banayein: Python_Class_1
3. Folder mein jayein
4. Folder ki contents dikhayein
5. Wapas previous folder mein aayein

Solution:

```
1 pwd # Location check
2 mkdir Python_Class_1 # Folder banayein
3 cd Python_Class_1 # Folder mein jayein
4 dir # Contents dikhayein
5 cd .. # Wapas jayein
```

4.3 Exercise 3: First .py Program

Exercise 3

Task: Cursor mein `hello.py` file banayein aur yeh program likhein:

```
Hello, I am learning Python
Python is easy
I will become an AI Engineer
```

Solution:

```
1 # hello.py
2 print("Hello, I am learning Python")
3 print("Python is easy")
4 print("I will become an AI Engineer")
```

4.4 Exercise 4: Homework Tasks

Homework - Class 1

Task 1: Brush karne ke 5 steps likhein (Roman Urdu/English)

Task 2: "Main AI Engineer kyun banana chahta/chahti hoon" - 3 sentences likhein

Homework - Class 2

Task:

1. Colab par new notebook banayein
2. 3 lines print karein: Name, City, Dream Job
3. Notebook ko Google Drive mein save karein

Homework - Class 3

Task:

1. Python install karein
2. `python --version` ka screenshot lein
3. Cursor install karein

5 Summary & Key Takeaways

Quick Reference Table

Concept	Key Point
Computer	Instructions follow karta hai
Python	Simple, beginner-friendly language
Interpreter	Line-by-line execute karta hai
Compiler	Saara code ek saath execute karta hai
REPL	Read-Eval-Print-Loop
.py File	Python program file
Google Colab	Online Python editor
Terminal	Text-based computer control

Important Points to Remember

1. **Case-sensitive:** Python mein capital/small letters ka farq hai
2. **No spaces:** Spaces ki jagah underscore (_) use karein
3. **.py extension:** Har Python file ka extension .py hona chahiye
4. **Add to PATH:** Installation ke waqt "Add to PATH" select karna mat bhoolen
5. **No AI yet:** Abhi AI features use nahi karna (Cursor ka AI off rakhein)
6. **Practice more:** Notes likhne se zyada coding karein

Course Progress Tracker

- ☐ Week 0 - Setup & First Run
- ☐ Week 1 - Variables & Data Types
- ☐ Week 2 - Control Structures
- ☐ Week 3 - Functions
- ☐ Week 4 - Lists & Tuples
- ☐ Week 5 - Dictionaries & Sets
- ☐ Week 6 - File Handling
- ☐ Week 7 - Error Handling
- ☐ Week 8 - OOP
- ☐ Week 9 - Testing & Debugging
- ☐ Week 10 - Data & API
- ☐ Week 11 - Async & First Agent
- ☐ Week 12 - Database & Deployment
- ☐ Week 13 - Capstone Project

6 Useful Resources

Download Links

- **Python:** <https://www.python.org/downloads/>
- **Google Colab:** <https://colab.research.google.com/>
- **Cursor:** <https://www.cursor.com/>
- **Python Documentation:** <https://docs.python.org/3/>

Quick Commands Reference

```
# Python Version Check
python --version

# Run Python File
python filename.py

# Open Python REPL
python

# Exit REPL
exit()

# Terminal Commands
pwd                # Current location
cd folder_name     # Change directory
cd ..              # Go back
dir                # List contents
mkdir folder_name  # Create folder
```